

Bathymetric and Side-Scan Data Fusion for Sea-Bottom 3D Mosaicing

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Abstract— The goal of sea-bottom mosaicing is to georeference each sample of a side-scan image in an accurate manner as this image consists only in a time-sampled backscattering recording. Thus, a flat sea-bottom assumption is generally used to spatially relocate each sample according to the height of the sensor. In the case of altitude varying sea-bottom (slopes, ridges, etc.), this hypothesis clearly introduces a bias in the process. Thus, we propose an algorithm that takes into account the altitude information, acquired by a multibeam echosounder, when available. Therefore, this data fusion results in a 3D mosaic, the shape of which corresponds to the topology of the sea-bottom while its textural aspect is provided by the side-scan images.

Keywords— Side-scan image, Bathymetry data, Mosaic.

INTRODUCTION

THE GOAL of sea-bottom mosaicing is to georeference each sample of a side-scan image in an accurate manner as this image consists only in a time-sampled backscattering recording. Thus, a flat sea-bottom assumption is generally used to spatially relocate each sample according to the height of the sensor (as provided by a user or determined automatically by tracking the water depth in the image). In the case of altitude varying sea-bottom (slopes, ridges, etc.), this hypothesis clearly introduces a bias in the process.

In this paper, the author presents a solution that takes into account, when available, the altitude sea-bottom information to finely position each side-scan sample at its real 3D location on the sea-bottom. Typically, one uses an acquisition system that supports both bathymetry and side-scan recording. This static configuration involves an accurate registration between the two sensors. Separately acquiring bathymetry and side-scan images requires an extra step to geographically match the two sets of information. Moreover, as the swath of these sensors can be different, a previous DTM of the area may also be used. The advantage of this data fusion representation is that it derives the benefit of each sensor's strength: Bathymetric information provides the geometric configuration of the sea-bottom, while side-scan images give a complementary

view of the sea-bottom reality (e.g., objects, sedimentology, fauna, flora indications, etc.)

Thus, the resulting mosaic can then be considered as a georeferenced 2D textural view of the sea-bottom which may be projected on the top of a DTM to produce a 3D view of the covered bottom. This rendering constitutes a first step and a realistic ground in an interactive virtual reality reconstruction of the sea environment.

This paper will use both KLEIN and RESON data acquired in the Sydney bay and part of the “Common Data Set” designed for the “Shallow Survey ’99” conference [1].

I. MOSAICING

Rough sonar images are given in the local reference system of the tow-fish, thus inducing some distortions when observing the same sea-bed region from different viewpoints (location, heading, and altitude), as illustrated in Figures 1 and 2.

Usually, sonar images are recorded with additional information about the tow fish configuration such as altitude (h), heading $g = (\alpha, \beta)$, location (φ, θ) , and, depending on the sonar system, beam spacing (p_b and p_e , at the beginning and at the end of the sonar range).

A. Basic projection

The goal of the projection process is to transfer all the data collected from the local time-varying tow-fish reference to a global georeferenced one, according to the navigational information provided by the tow-fish.

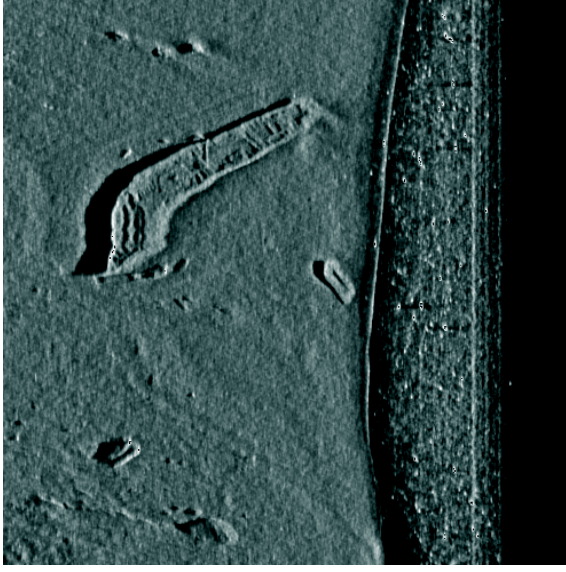
We note $\Im$, the function which returns the real distance SS' on the sea-bed for a point I in the image. Assuming $\Re(bed_dist)$ gives the altitude of the point S' located at the distance bed_dist from S in a direction $g^\perp$, $bed_dist = \Im(obl_dist)$ verifies the following equation.

$$\Im(obl_dist) = \sqrt{obl_dist^2 - (\Re(bed_dist) - h)^2} \quad (1)$$

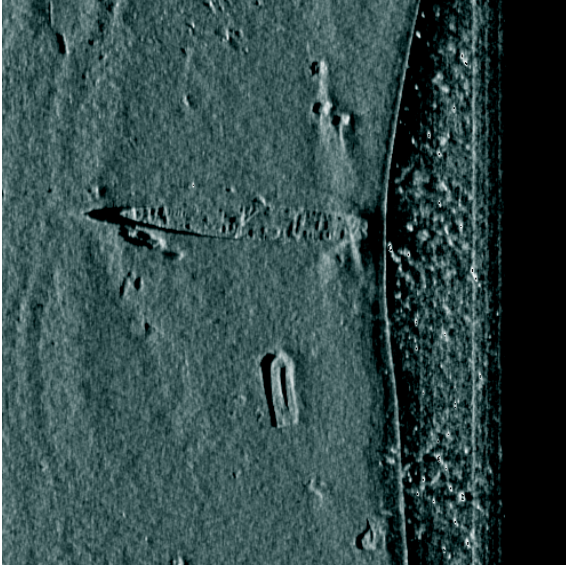
In case of the flatness of the sea-bottom due to the lack of the relief information, $\Im$ has the following expression.

$$bed_dist = \Im(obl_dist) = \sqrt{obl_dist^2 - h^2} \quad (2)$$

We define the function Ψ which returns the coordinates of the point $S'(\varphi', \theta')$, given the point $S(\varphi, \theta)$, the distance $d = SS'$ and the perpendicular normalized vector



(a) Track 1



(b) Track 2

Fig. 1. Raw sonar images from different tracks (KLEIN 5000)

$g = (\alpha, \beta)$ to (SS') .

$$S' = \Psi(S, d, g) \text{ with } \begin{cases} \varphi' &= \varphi \mp |\beta d| \\ \theta' &= \theta \pm |\alpha d| \end{cases} \quad (3)$$

Thus, for each pixel (i, j) of a recurrence, the vertices of the projected polygon C have the following coordinates.

$$C_{(i,j)} = \Psi(S_i, p_{i,j}, g^\perp) \quad (4)$$

$$C_{(i+1,j)} = \Psi(S_{i+1}, p_{i+1,j}, g^\perp) \quad (5)$$

$$C_{(i,j+1)} = \Psi(S_i, p_{i,j+1}, g^\perp) \quad (6)$$

$$C_{(i+1,j+1)} = \Psi(S_{i+1}, p_{i+1,j+1}, g^\perp) \quad (7)$$

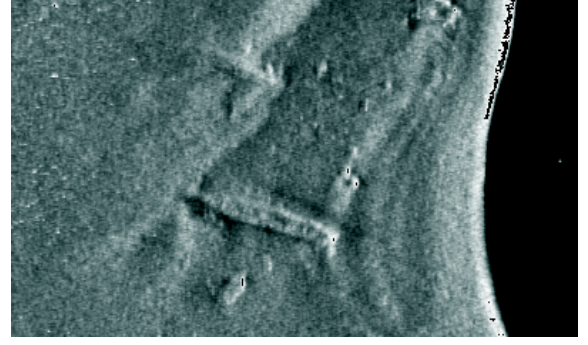


Fig. 2. Raw sonar image from SeaBat 8101

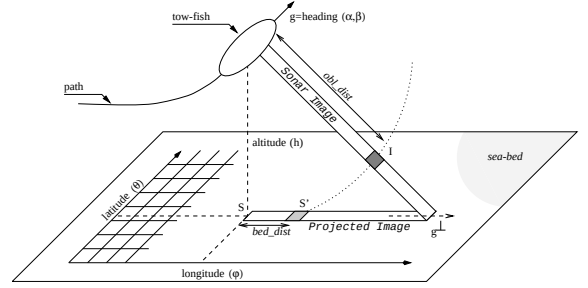


Fig. 3. Sonar in-situ schematic representation

where

$$S_i = \Psi(S, d_i, g) \quad (8)$$

$$d_i = \Im(pixel_resolution * i) \quad (9)$$

$$p_{i,j} = \frac{(p_d + i * p_e - p_d I)}{J} * (j - \frac{J}{2}) \quad (10)$$

$$(i, j) \in [0, I - 1] \times [0, J - 1] \quad (11)$$

B. Integrating the environment

Three major topics clearly participate in the geo-referencing of the side-scan sonar images.

- *image formation rules*: each sonar system has specific methods when building the raw images [2].
- *sound velocity profile*: the variations of the sound velocity with depth greatly influence the real shape of the insonified area [3].
- *relief*: the flat bottom assumption [4] induces errors in the relocation of the image samples as illustrated in Figure 4.

Some of these issues have been addressed in a simulator based on ray-tracing methods [5]. Nevertheless, the computational cost of such a method “From sensor to bottom” is very important and increases with the amount of samples in the range as each sample needs to be relocated to form the mosaic.

The proposed approach, “From bottom to sensor,” reverses the situation by tailoring the computation burden to the requested resolution for the mosaic.

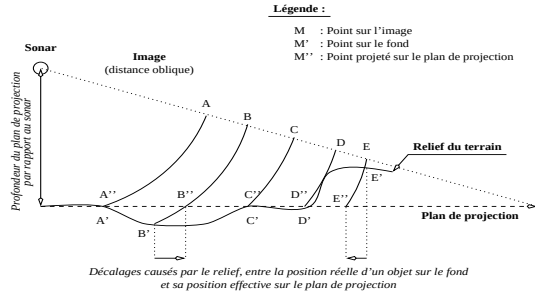


Fig. 4. Errors due to the flat sea-bed assumption

II. 3D MOSAICING

As we want to integrate various information coming from different sensors, some registration procedures need to be performed when the relative location of all the used sensors is either not perfectly known or time-varying (for instance, when dealing with two separate acquisitions).

In this study, the concerned sensors are an MBES¹ (for the bathymetry data) and an SLS² (for the images). Thus, as the intrinsic basis of the data to be matched are not similar, a symbolic matching process is preferred. This consists of first extracting some *symbolic* features from both sets of data and then, if they represent the same topological features, to mate them in order to create a geographic anchor between the two datasets [6]. Registration algorithms between information coming from several sensors or tracks must cope with two types of problems: segmentation errors resulting from the extraction processes and navigational uncertainties resulting from technological limitations [7].

The RESON system of acquiring both bathymetry and images simultaneously leads to a straightforward registration algorithm as the offsets between the two sensors are static (and thus not time-varying).

A. Sea-bottom topology

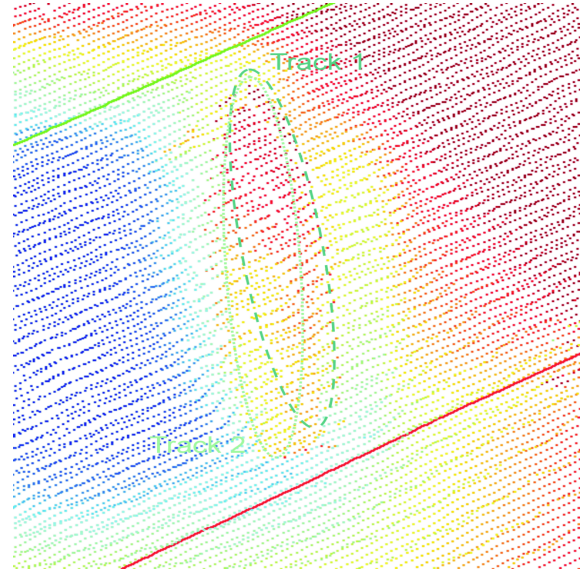
Even if bathymetry and images are acquired simultaneously, their swath may not be identical. To solve this problem, we chose to build a topological representation (DTM) of the sea-bottom by gathering all the bathymetry data acquired during the survey. This DTM will then feed the mosaicing process.

A.1 Bathymetry calibration

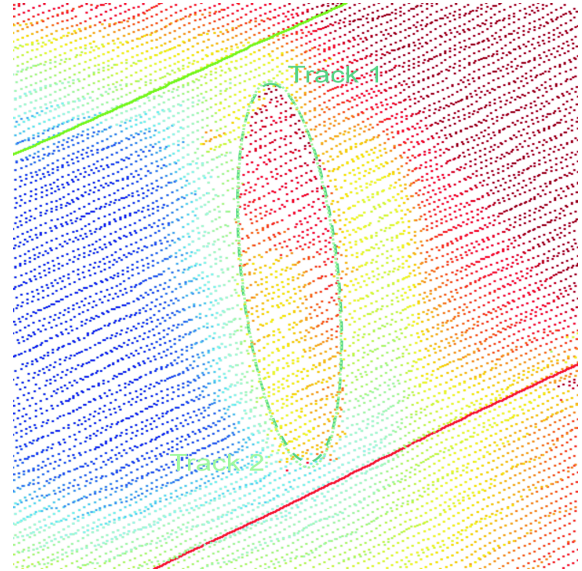
A calibration step is crucial for all bathymetry surveys. We designed a “Patch Test” program [8], the goal of which is to precisely determine the static configuration of the sonar head (roll, pitch, yaw) and the latency remaining between the reception of the GPS fix and its integration by the acquisition system. An imprecise knowledge of

¹Multibeam echosounder

²Side-scan looking sonar



(a) Raw XYZ data



(b) After calibration

Fig. 5. Impact of the calibration procedure on the XYZ data

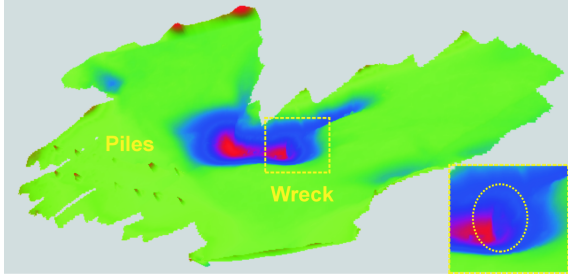
these four values introduces bias when computing the real 3D geographic location (XYZ) of the intersection of each beam with the sea-bottom.

Figure 5 shows the resulting XYZ coming from two tracks on the wreck area, with and without calibration. The offsets are $\{\text{roll} = -0.1^\circ, \text{pitch} = -1.0^\circ, \text{yaw} = 2.5^\circ\}$ while the latency is estimated to be 180ms. The contour of this wreck is outlined on both the tracks before and after the patch test procedure.

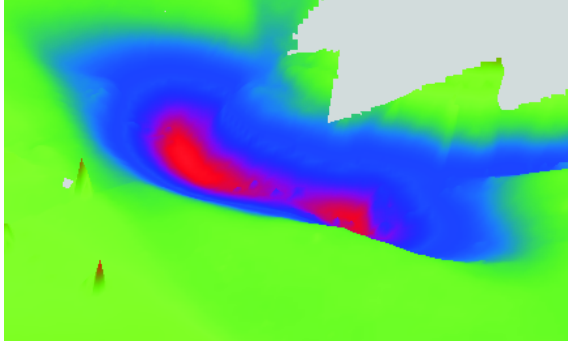
A.2 Building the DTM

The DTM is defined as an altitude-oriented grid which geographic location and resolution need to be defined according to the characteristics of the MBES used. An aggregation operator is then required to merge the XYZ data belonging to the same cell of the grid.

Figure 6 presents an example of a DTM built from the bathymetry data coming from the Western survey [1]. Twenty tracks were merged to obtain this DTM, which has a resolution of 50cm x 50cm. The DTM presents some interesting features, like piles from a bridge and some wrecks.



(a) Global DTM



(b) Zoom on the wreck area

Fig. 6. Example of computed DTM

B. Sidescan projection for mosaicing

The basic method of projection involves the use of a processing unit for each sample of the range in order to determine its covered area on the sea-bottom. In such a case, the complexity of the mosaicing process is directly linked to the resolution of the sensor and is not affected by the requested mosaic resolution.

The new approach follows the inverse path by computing the contribution from the samples to each pixel of the mosaic. Thus, the computation burden only depends on the asked resolution of the final mosaic.

B.1 From the bottom to the sensor

The following algorithm is the elementary unit required to compute the value of the pixel $P_{i,j}$ in the mosaic for

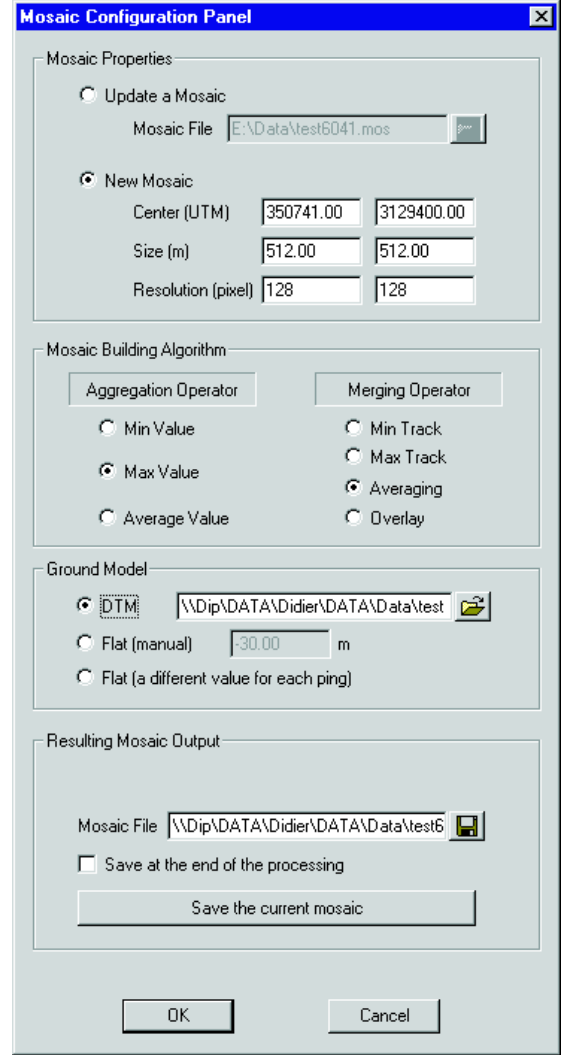


Fig. 7. Building a mosaic

one considered ping.

For this considered ping, V is the volume insonified by the acoustic wave. V can be determined by a mathematical formula but is more realistically defined by performing a procedural routine that takes into account current extra information such as the system dynamic attitude (roll, pitch, and yaw) and the current sound velocity profile.

Each pixel $P_{i,j}$ of the mosaic is defined by the polygon $C = \langle C_{(i,j)}, C_{(i+1,j)}, C_{(i+1,j+1)}, C_{(i,j+1)} \rangle$, where each $C_{(k,l)}$ is a 3D vertex. Its X- and Y-components directly come from the mosaic while the Z-component results from the supporting DTM. The next step will determine which samples are concerned, within the ping, with the pixel $P_{i,j}$.

- The minimum (resp. maximum) distance d_{min} (resp. d_{max}) between the sensor and C is estimated.
- This value induces the duration t_{min} (resp. t_{max}) for the sound to perform the two-way travel from the sensor to the sea-bottom, at the location of d_{min} (resp. d_{max}).

- Corresponding to the duration t_{min} (resp. t_{max}), the sample a (resp. b) in the SLS range, is then determined utilizing the sound velocity characteristics.

If $V \cap C \neq \emptyset$ then $P_{i,j} = \bigoplus_{k \in [a,b]} R(k)$ where $\bigoplus$ is the *pixel aggregation operator*. As shown in Figure 7, three operators are currently available: minimum, maximum, and average.

B.2 Track overlapping

A survey usually consists of a series of tracks which partially overlap. Building a mosaic implies a two-step processing to merge the different projected tracks (local mosaics) inside one global mosaic.

Depending on the accuracy of the localization, a matching procedure may be needed to symbolically anchor the relative mosaics using reliable features occurring on them. This step makes it possible to relocate each mosaic in a global reference system through a set of geometrical transformations [7]. The set may be the identity transformation when localization is accurate enough.

Even after these transformations are known, the local mosaics still provide different “images” of the sea-bottom. Due to the sonar acquisition mechanism, their aspects can be very different, especially around altitude-varying features. For example, looking at the wreck from a north-bound or an eastbound track will produce two different shadows on the mosaics. A user decision is required in order to perform the final pixel-intensity merging. As shown in Figure 7, four *track merging operators* are defined according to the final use of the resulting mosaic:

- minimum: preserve the different viewed shadows
- maximum: focus on the different echo structures
- average: compromise for a global view
- last track: the last track is the preserved one

III. RESULTS

A. Mosaic Output

Figure 8 presents a 2D view of the produced mosaic, on the wreck area, using RESON data. As this view is geo-referenced, this mosaic can be overlapped onto the DTM of the region to build a 3D representation of the environment. The shape of this 3D mosaic corresponds to the topology of the sea-bottom while the textural aspect is provided by the side-scan images, as shown in Figure 9.

B. Multi-scale representation

This program can build various mosaics of the same sea-bottom at different resolutions, feeding 3D rendering tools that support multi-scale representation through level of details mechanisms. It means that the adequate resolution of the mosaic to be displayed will be selected in the tree of mosaics according to the distance between the viewpoint and the surface to render. Thus, we computed such a multi-resolution tree of mosaics with three levels and used a VRML-compatible browser as a display.

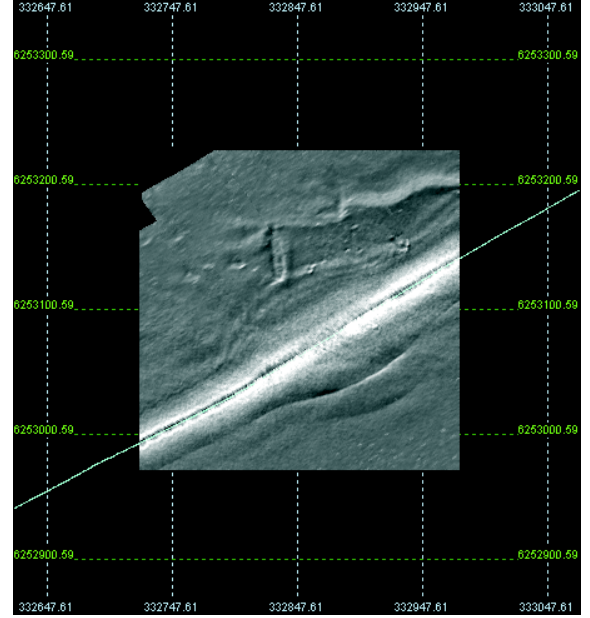


Fig. 8. 2D View of the area

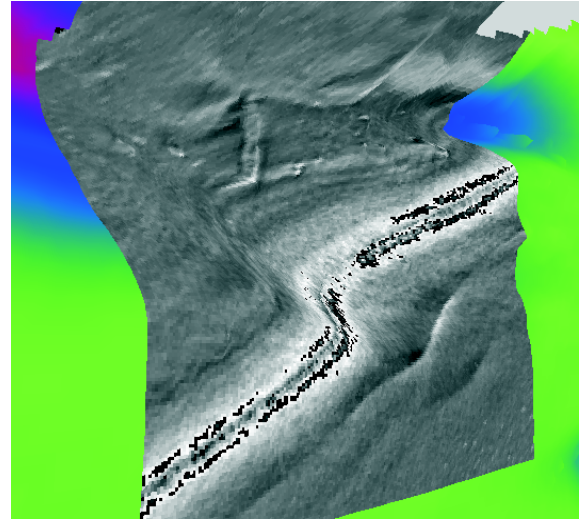
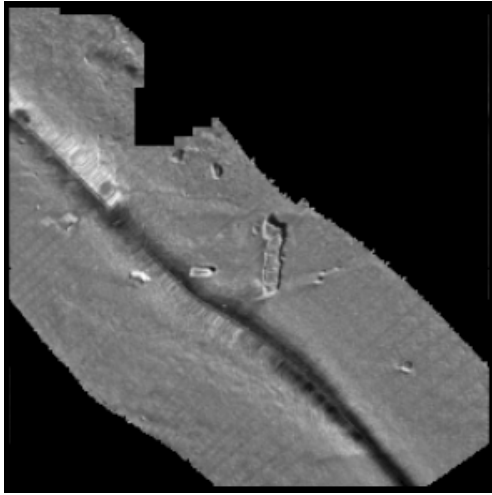


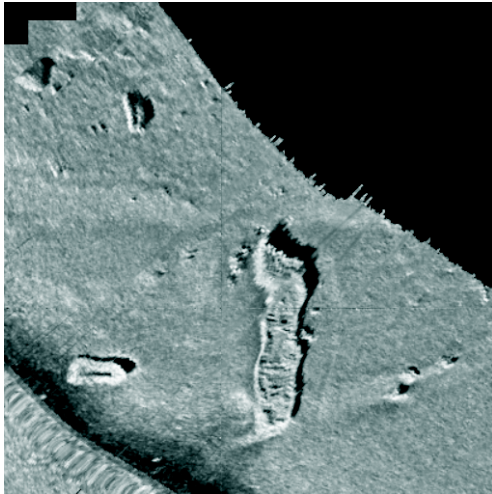
Fig. 9. Zoomed 3D view of the wreck area

Figure 10 shows levels 1 and 3 built for the area of the wreck. Each level η covers the same area with $2^{\eta-1} \times 2^{\eta-1}$ elements. Each element is a 256x256 pixel mosaic, so a level-3 element has four times the resolution of the level-1 element, as its covered surface is four times smaller.

Working with multi-resolution trees allows building small, high-resolution mosaics which focus on the interesting location. Dealing with a one-level large mosaic would increase the computation cost, while degrading the resolution would result in fuzzy mosaics. This latter case is demonstrated in Figure 11 where a level-1 mosaic is directly zoomed to simulate the same viewpoint as the one used for the level-3 mosaic in Figure 10(b).



(a) Level 1 (1 element)



(b) Level 3 (4x4 elements)

Fig. 10. Examples of multi-resolution tree of mosaics

FURTHER DEVELOPMENTS & APPLICATIONS

This first step towards a complete representation of the sea-bottom through virtual reality integrates two fundamental methods for acquiring knowledge of the sea-bottom: sonar images and bathymetry data. Building such a 3D mosaic by merging this information represents the first step in gathering multiple sensor outputs to represent the acquisition environment, thus offering a virtual reality of the underwater scene.

This 3D environment can also come from previous acquisitions stored in databases; through proper symbolic registration mechanisms, this virtual world enables an Underwater Unmanned Vehicle autopilot to estimate the current location of the vehicle, and to coherently update its databases.

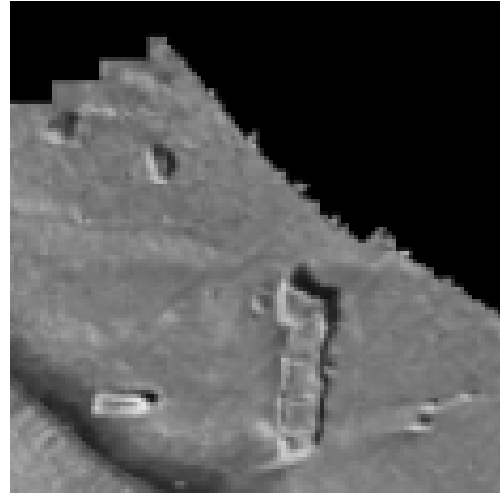


Fig. 11. Zoom on the level 1 mosaic to simulate a level 3 rendering

REFERENCES

- [1] DSTO Australia Defense Science & Technology Organization, "Common data set," in *Shallow Survey 99 - International Conference on High Resolution Surveys in Shallow Water*, Sydney, Australia, October 1999.
- [2] Didier Guériot, Julien Chèdru, and Eric P. Maillard, "A new tracking technique for sonar images relying on symbolic information matching through genetic algorithms," in *IASTED International Conference Signal and Image Processing*, New Orleans, December 1997, pp. 353–358.
- [3] Sylvie Daniel, Didier Guériot, and Eric P. Maillard, "Underwater environment restitution through sonar images and swath bathymetry rendering," in *IEEE Conf. on Systems, Man and Cybernetics*, October 1998.
- [4] Didier Guériot, *Utilisation des algorithmes génétiques pour des problèmes d'optimisation spécifiques - Application aux réseaux de neurones et au traitement d'images sonar*, Ph.D. thesis, Université de Haute Alsace, Décembre 1998.
- [5] Didier Guériot, Sylvie Daniel, and Eric P. Maillard, "Sonar images and swath bathymetry simulations for environment restitution through acoustical ray tracing techniques," in *Fourth European Conference on Underwater Acoustics*, Rome, Italy, September 1998, vol. 1, pp. 425–431.
- [6] Didier Guériot, Eric P. Maillard, and Benoit J. Zerr, "Sonar image registration by symbolic matching using genetic algorithms," in *Underwater Defense Technology*, Cannes, July 1995, pp. 239–243.
- [7] Didier Guériot, Eric P. Maillard, and Jean-Pierre Kernin, "Sonar image registration through symbolic matching: a fuzzy local approach using genetic algorithms," in *Oceans MTS / IEEE*, Fort Lauderdale, September 1996, vol. 3, pp. 1324–1329.
- [8] Didier Guériot, Julien Chèdru, Sylvie Daniel, and Eric Maillard, "The patch test: A comprehensive calibration tool for multibeam echosounders," in *OCEANS 2000 MTS/IEEE*, Providence, Rhode Island, September 2000.